DataHunt

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Contents

```r
#Easy Question
n <- 30
x <- 6
p0 <- 0.10
alpha <- 0.05
```

```r
p_hat <- x / n
p_value <- pbinom(x - 1, size = n, prob = p0, lower.tail = FALSE)

se <- sqrt(p_hat * (1 - p_hat) / n)
ci_low  <- p_hat - 1.96 * se
ci_high <- p_hat + 1.96 * se
```

```r
cat("Sample rate:", p_hat, "\n")
```

```
## Sample rate: 0.2
```

```r
cat("p value:", p_value, "\n")
```

```
## p value: 0.07319011
```

```r
cat("95% CI: (", ci_low, ",", ci_high, ")\n")
```

```
## 95% CI: ( 0.05686184 , 0.3431382 )
```

The p value of 0.073 exceeds alpha = 0.05, so we fail to reject H_O and do not have sufficient evidence to conclude the defect rate exceeds 10%. The 95% CI (5.7%, 34.3%) is wide given n = 30, meaning the data are consistent with the factory's claim but the sample is too small to be conclusive!

```r
#Medium Question
#Hypotheses
#H_0: mu_engg = mu_science (like the mean coffee consumption is equal across both groups)
#H_1: mu_engg > mu_science (that means engineering students drink more coffee)
#one sided question because it asks if engg students drinks more?

#TEST imma use
#Two independent groups, continuous outcome, n = 50 /group. So a two sample
#t test (Welch's probably, which does not assume equal variances). I might even check
#normality Shapiro Wilk.
```

```r
df <- read.csv("data/students_and_coffee.csv")
df$faculty <- trimws(df$faculty) #THE DEBUGGING OMG!!!!!

eng <- df$coffee_cups_per_day[df$faculty == "Engineering"]
sci <- df$coffee_cups_per_day[df$faculty == "Science"]
```

```r
mean(eng)
```

```
## [1] 3.02
```

```r
mean(sci)
```

```
## [1] 2.46
```

```r
shapiro.test(eng)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  eng
## W = 0.94573, p-value = 0.02285
```

```r
shapiro.test(sci)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  sci
## W = 0.96447, p-value = 0.1368
```

```r
#actual welch two sample t test
t.test(eng, sci, alternative = "greater", var.equal = FALSE)
```

```
##
##  Welch Two Sample t-test
##
## data:  eng and sci
## t = 2.8993, df = 97.654, p-value = 0.002309
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
##  0.2392571       Inf
## sample estimates:
## mean of x mean of y
##      3.02      2.46
```

The engg group's Shapiro Wilk test returns p = 0.023, which technically rejects the normality. However, with n = 50 the t test is still robust to mild departures from normality by the CLT, so the Welch t test result do remains valid.

The Welch two sample t test yields t = 2.90 on df = 97.65, with a one sided p val of 0.0023. Since p val<alpha = 0.05, we reject the null hypothesis. There is statistically significant evidence that engg students consume more coffee per day than science students on average (3.02 vs 2.46 cups/day), with the 95% CI for the difference bounded below at 0.24 cups. This would be my final conclusion!

```r
#HARD STUDENT PERFORMANCE INSIGHT
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.2
```

```r
df <- read.csv("data/student_performance_insights.csv")
```

```r
df$Stress_Numeric <- as.numeric(factor(df$Stress_Level,levels = c("Low", "Moderate", "High")))
```

```r
#imma use multiple linear regression model
model <- lm(GPA ~ Study_Hours_Per_Day + Sleep_Hours_Per_Day +
                  Social_Hours_Per_Day + Physical_Activity_Hours_Per_Day +
```

```r
                  Extracurricular_Hours_Per_Day + Stress_Numeric, data = df)

summary(model)
```

```
##
## Call:
## lm(formula = GPA ~ Study_Hours_Per_Day + Sleep_Hours_Per_Day +
##     Social_Hours_Per_Day + Physical_Activity_Hours_Per_Day +
##     Extracurricular_Hours_Per_Day + Stress_Numeric, data = df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.5944 -0.1352 -0.0030  0.1344  0.7874
##
## Coefficients: (1 not defined because of singularities)
##                                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)                     1.8272509  0.0929090  19.667   <2e-16 ***
## Study_Hours_Per_Day             0.1618014  0.0064501  25.085   <2e-16 ***
## Sleep_Hours_Per_Day             0.0029802  0.0052889   0.563   0.5732
## Social_Hours_Per_Day            0.0088093  0.0045097   1.953   0.0509 .
## Physical_Activity_Hours_Per_Day 0.0074958  0.0039608   1.892   0.0586 .
## Extracurricular_Hours_Per_Day          NA         NA      NA       NA
## Stress_Numeric                  0.0002071  0.0104942   0.020   0.9843
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2026 on 1994 degrees of freedom
## Multiple R-squared:  0.541,  Adjusted R-squared:  0.5398
## F-statistic:   470 on 5 and 1994 DF,  p-value: < 2.2e-16
```

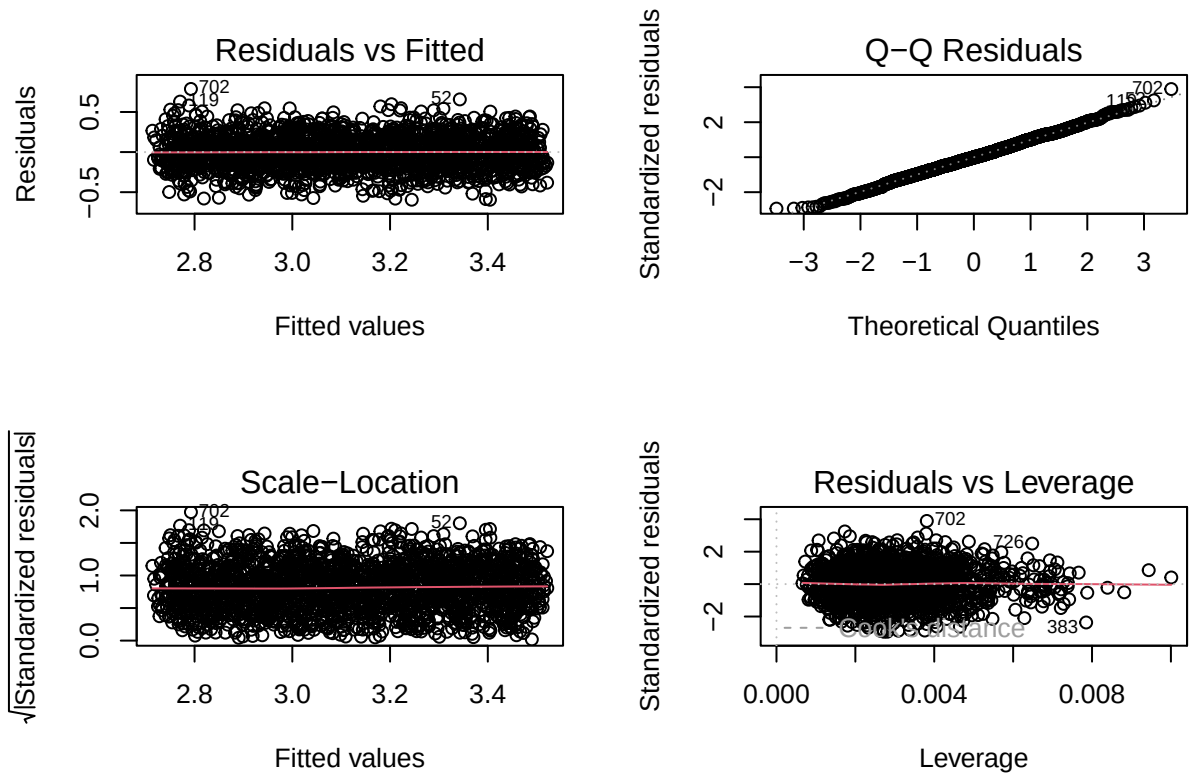
```r
#scale it because i want to compare the var importance
df_scaled <- as.data.frame(scale(df[, c("GPA", "Study_Hours_Per_Day",
                                        "Sleep_Hours_Per_Day","Social_Hours_Per_Day",
                                        "Physical_Activity_Hours_Per_Day",
                                        "Extracurricular_Hours_Per_Day",
                                        "Stress_Numeric")]))
```

```r
model_std <- lm(GPA ~ ., data = df_scaled)
summary(model_std)
```

```
##
## Call:
## lm(formula = GPA ~ ., data = df_scaled)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.99020 -0.45252 -0.01005  0.44988  2.63626
##
## Coefficients: (1 not defined because of singularities)
##                                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)                     8.349e-15  1.517e-02   0.000   1.0000
## Study_Hours_Per_Day             7.714e-01  3.075e-02  25.085   <2e-16 ***
## Sleep_Hours_Per_Day             1.458e-02  2.587e-02   0.563   0.5732
## Social_Hours_Per_Day            4.980e-02  2.550e-02   1.953   0.0509 .
## Physical_Activity_Hours_Per_Day 6.310e-02  3.334e-02   1.892   0.0586 .
```

```
## Extracurricular_Hours_Per_Day          NA         NA      NA       NA
## Stress_Numeric                  5.044e-04  2.556e-02   0.020   0.9843
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6784 on 1994 degrees of freedom
## Multiple R-squared:  0.541,  Adjusted R-squared:  0.5398
## F-statistic:   470 on 5 and 1994 DF,  p-value: < 2.2e-16
```

```r
par(mfrow = c(2, 2)) #better graphics incase its new to yall
plot(model)
```



MODEL: This is my multiple linear regression predicting GPA from study hours, sleep, social hours, physical activity, extracurricular hours, and stress level. n = 2000, adjusted r_squared = 0.54. Extracurricular_Hours_Per_Day is automatically dropped due to perfect multicollinearity with the remaining predictors combined okay. It has no independent info on its own.

FACTORS: Study hours is the only statistically significant predictor (p very small). Its std beta of 0.771 means it accounts for the overwhelming majority of explained variance. No other variable literall even comes close. Sleep, social hours, physical activity, and stress all have standardised betas below 0.07 and FAIL to reach significance at alpha = 0.05. For practical purposes, this model reduces to: GPA is largely a function of how many hours a student studies per day.

INTERPRET- Each additional hour of study per day is associated with a GPA increase of approX 0.16 points, holding all other vars constant. The model explains 54% of GPA variance, which is reasonable for observational lifestyle data in my opinion tbh. The diagnostic plots support the model assumptions: residuals vs fitted shows no sys curvature, the QQ plot tracks the diagonal closely indicating approx normality, scale location shows roughly constant spread :), and no points breach Cook's distance thresholds in the leverage

plot. YAYYYY Observations 702, 520, and 383 are flagged as mild outliers but do not appear to be distorting the fit given the large sample size.

LIMITATION - The dataset contains no measure of prior academic ability or even course difficulty bruh. Study hours almost certainly captures some of the effect of student aptitude, but since high ability students may study more and also earn kinda higher GPAs independently (you know what i'm talking about). This produces sorta an upward bias in the study hours coeff. The true causal effect of an add. hour of study is likely smaller than 0.16 GPA points. Without an ability control variable, the coefficient cannot be interpreted causally. I think this would be the biggest limitation here.